Steven Koehl

Software Engineer | Aerospace & Avionics | AI & Workflow Automation

Fairchild AFB, WA · contact@stevenkoehl.com · (567) 201-6985 · stevenkoehl.com · linkedin.com/in/steven-koehl · github.com/sleuthy-sloth

Active Secret Clearance · SkillBridge-eligible Nov 2027

PROFESSIONAL SUMMARY

U.S. Air Force Technical Sergeant retiring June 2028 after 18 years across avionics/circuit-level maintenance, technical program management, and quality engineering. Managed a $35M fleet/equipment portfolio, a $2M service contract at 100% PWS adherence, and 140 PQDRs recovering $7M. Holds an M.S. in Aeronautics and B.S. in Aeronautical Engineering from Embry-Riddle Aeronautical University. Builds production software on the side in C++, TypeScript/React, and Python. Seeking software engineering roles in defense/aerospace where operational domain knowledge is an asset.

CORE COMPETENCIES

Languages: C++20, TypeScript, JavaScript, Python, SQL

Frontend: React 19, Next.js, Tailwind CSS, PWA (Workbox/IndexedDB), Vite

Backend & Data: FastAPI, Firebase (Auth/Firestore/App Check), SQLite, REST, SSE

Desktop & Systems: Qt 6, OpenGL/VTK, DCMTK, CMake, vcpkg

AI & Automation: Multi-provider LLM integration, OCR/document parsing, prompt engineering, workflow automation

DevOps: GitHub Actions, Docker, CI test gates, Codeberg mirror, GitHub Pages

Domain: Avionics systems, circuit-level diagnostics, Technical Orders, JDRS/AFMC liaison, COR oversight, ISD

PROFESSIONAL EXPERIENCE

Senior AFREP Technician & AI Developmental Lead — U.S. Air Force, Fairchild AFB, WA

Oct 2024 – Present

- Senior technician in the Air Force Repair Enhancement Program: micro-miniature circuit-card repair, component-level maintenance, and equipment sustainment; local repair of high-cost NMC assets to improve wing mission capability rates.
- Lead evaluation and adoption of AI tooling for aerospace maintenance workflows (document OCR, intelligence feeds, form-to-data extraction).
- Coordinate Technical Order distribution, product improvement activities, and service-contract oversight across maintenance programs.

Maintenance Program Manager — Technical Order Distribution, Product Improvement & COR — U.S. Air Force, JB McGuire-Dix-Lakehurst, NJ

Jan 2020 – Oct 2024

- Orchestrated a tri-functional management strategy (TODO + PIM + COR) ensuring mission readiness for a fleet/equipment portfolio valued at $35M.
- Managed a library of 5,000 digital/physical Technical Orders with monthly audits at a 98% accuracy rate; ensured all maintenance actions met Air Force/FAA-equivalent safety standards.
- Primary AFMC liaison for the Joint Deficiency Reporting System (JDRS): investigated and submitted 140 PQDRs, recovering $7M and driving fleet-wide safety modifications.
- Contracting Officer Representative for a $2M service contract: 100% PWS adherence, certified monthly invoices, vendor performance surveillance.
- Leveraged deficiency data to identify training gaps; revised the TO distribution process, reducing maintenance errors by 15%.
- Conducted contractor/technical-data surveillance; enforced OSHA and AFOSH compliance, eliminating safety hazards.

Master Technical Trainer & Instructional System Designer — U.S. Air Force, Sheppard AFB, TX

Jan 2016 – Dec 2019

- Subject Matter Expert for a 200-student annual pipeline; authored/revised 300 hours of instructional material integrating ETIMS (Technical Orders) and AFREP repair scenarios.
- Delivered 4,000+ hours of platform instruction on complex electronics, micro-miniature soldering, and component-level diagnostics; maintained a 92% student pass rate.
- Managed daily schedules, discipline, and academic progress for an 800-student pipeline; 100% on-time graduation for high-priority AFSCs.
- Developed written and performance-based evaluations; data-driven analysis raised class averages by 12%.

- Enforced OSHA/AFOSH in high-risk labs with zero training mishaps; served as MTL-equivalent for professional development and career counseling.

Avionics Technician — Circuit-Level Maintenance — U.S. Air Force
2008 – 2016

- Circuit-level troubleshooting, diagnostics, and repair on aircraft avionics and electronic systems; micro-miniature circuit-card repair, soldering, component analysis.
- Supported aerospace maintenance operations through technical problem-solving and equipment sustainment.

SELECTED PROJECTS

Clivus — Cross-Platform DICOM Viewer · C++20, Qt 6.8, DCMTK, OpenGL, VTK, SQLite, CMake
github.com/sleuthy-sloth/clivus

- Open-source (GPL-3.0) Windows-first DICOM viewer built test-first; CI runs an assert-based fidelity suite on Windows (core + GUI builds).
- Window/level re-render of a 512^2 frame in 0.34 ms against a 16.7 ms frame budget (~2 orders of magnitude headroom), enabling drag-to-window interaction on 12-bit CT series.
- ROI measurements (mean/SD/min/max/area mm^2) saved per series and slice and exportable to CSV; MPR reformats and DICOMweb (QIDO-RS/WADO-RS) study loading.
- Offline by construction (no network, telemetry, or accounts); SQLite series index cuts folder reopen from ~30 ms to ~1.5 ms.

92 AMXS Tracker — Squadron Maintenance Command Center · React 19, TypeScript, Tailwind 4, Firebase, PWA
github.com/sleuthy-sloth/92-AMXS-Tracker · live: sleuthy-sloth.github.io/92-AMXS-Tracker

- Production maintenance/training/logistics platform for the 92nd Aircraft Maintenance Squadron: maintenance logs, DIFM parts pipeline, training readiness, Red Ball alerts, shift-turnover PDFs.
- 16 pages with role-based access (NCOIC/leadership views); offline-capable PWA via Workbox + IndexedDB.
- Dual-provider AI layer (DoD GenAI.mil primary → OpenRouter fallback) with Zod-validated responses; AI OCR extracts tail/JCN/discrepancy/repair from photos of AF Forms 781A/781K; loop-closure detection flags recurring discrepancies on a tail.
- Deployed via test-gated GitHub Actions with a Codeberg mirror; ~170 lines of Firestore security rules.

OpenNarrator — Local Audiobook Factory · Python 3.11, FastAPI, Typer, Kokoro TTS, ffmpeg
github.com/sleuthy-sloth/opennarrator

- Converts EPUB/TXT/DOCX into chaptered M4B audiobooks entirely on-device (no API keys); web UI + CLI; 10 voices; per-format chapter detection.
- 85 unit tests; CI on GitHub Actions; packaged for pip, Homebrew, Docker, and a native desktop app; GitHub Actions audiobook-factory template.
- RTF 0.16× on Apple MPS (~1.6 h for a 10-hour book); loudnorm + concat + AAC packaging with chapter markers and cover art.

EDUCATION

M.S. Aeronautics — Embry-Riddle Aeronautical University Worldwide *Jun 2019 – Jun 2021*

B.S. Aeronautical Engineering — Embry-Riddle Aeronautical University Worldwide *Aug 2016 – May 2019*

Instructor of Technology and Military Science — Community College of the Air Force, Maxwell AFB *Jan 2016 – Jun 2018*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Active Secret clearance
- Contracting Officer Representative (COR) oversight
- Instructional System Design (ISD)
- Joint Deficiency Reporting System (JDRS) / AFMC liaison
- AI integration and workflow automation initiatives

ADDITIONAL

- Languages: English (native), Spanish (highly proficient), French
- Open-source contributor; homebrew-tap and Codeberg mirror workflows across projects